

Eleven exactly certified violations of the proofreading bound $e^{-2\Delta}$ among 88 two-bound-state networks

Verifier-Gated Discovery Lab

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Abstract

Hopfield’s kinetic proofreading scheme [1] has the limiting error rate $e^{-(n+1)\Delta}$ for a linear chain with n proofreading steps. Whether this bound carries over to networks with two bound states is open: the classification of the enumerated two-bound-state family was not found in a targeted literature search. We fix $\Delta = \ln 100$ and classify the 88 non-degenerate rule-compliant two-bound-state topologies. The bound $\eta \geq e^{-2\Delta}$ is proven for 50 of these topologies. It is exactly violated for 11, with $\eta < 1\text{e-}4$ and values down to $1.386398\text{e-}12$. For the remaining 27 it is open. For the linear chain with $n = 2$, $\eta \geq e^{-3\Delta}$ holds for all rates. Bounds are proved by symbolic positivity certificates; violations are certified with exact rational rates. The main limitation is that 27 topologies remain undecided. All violation certificates lie in a bounded log-rate range.

Keywords: kinetic proofreading, Hopfield bound, error rate, entropy production, Markov networks, exact certificates

1 Introduction

Kinetic proofreading raises specificity beyond what free-energy differences or kinetic barriers give, when the reaction is driven strongly but nonspecifically [1]. Error, speed and dissipation trade off along Pareto fronts, studied analytically and numerically for energy-relay proofreading [2], generalized Hopfield models [3], and master-equation frameworks [4]. Error bounds fixed by transition-state energies have been derived for proofreading networks [5, 6]. Several checking steps can reach a given accuracy at lower dissipation than a single one [7]. Scaling analyses exist for Hopfield’s original network [8]. Discrimination regimes can include anti-proofreading [9], and excessive discrimination can remove the nonequilibrium advantage [9].

Open question. Does $\eta \geq e^{-2\Delta}$ hold for the enumerated two-bound-state topologies, or can topologies other than the linear chain with one proofreading step go below it?

Contribution. We classify the whole enumerated family with two bound states into proven, certified-violated and open topologies. The classification uses exact certificates only. A numerical optimiser serves as a candidate generator.

Summary of results.

- Theorem 1 shows the bound $\eta \geq e^{-3\Delta}$ for the linear chain with $n = 2$, for all rates and fuels.
- Theorem 2 gives a symbolic proof of $\eta \geq 1/D^2$ for 50 topologies.
- The next three theorems give exact violations for fam2_66, then for fam2_11 and fam2_9, then for ten topologies in total.
- The final two theorems give the complete classification (50 / 11 / 27) and the table of exact error rates.
- Propositions and Examples give supporting statements and exact reachable points for the linear chain.

2 Model and assumptions

All networks follow the rules in. The fixed parameters are:

Parameter	Value	Meaning
Δ	$\ln 100$ ($e^\Delta = 100$, $e^{-\Delta} = 0.01$)	discrimination free-energy difference in kT ; W leaves bound states e^Δ times faster
L	10	every rate, including derived reverse rates, lies in $[e^{-10}, e^{10}]$
μ	≥ 0 , explored in $[0, 20]$	fuel chemical potential (kT) per fuel-driven step
μ_P	≥ 0 , explored in $[0, 20]$	chemical potential (kT) of product formation
max_states	10	maximal number of states of a network
concentrations	1	substrate concentrations absorbed into binding rates
rationalisation	denominator $\leq 10^6$	rates rounded to rationals for exact certificates
family_k	2	number of bound states in the enumerated family

Notation. $D = e^\Delta$ and $G = e^\mu$. The error rate is $\eta = J_W/J_R$, the speed is $v = J_R$, and σ is the entropy production per product in units of kT . These are fixed conventions of the problem statement.

Family. The verifier’s generator produces 92 rule-compliant two-bound-state topologies. Four are degenerate (no net production, $\eta = 0/0$) and are excluded, leaving 88.

3 Method

Certificate (a), exact reachability. Rational rates (denominator $\leq 10^6$, log-rates in $[-10, 10]$, local detailed balance) are inserted into the steady-state equations of a topology. η , σ and v are evaluated in exact rational arithmetic and compared with the threshold.

Certificate (b), symbolic positivity. A bound $\eta \geq B$ is proved by writing $\eta - B = N/D$ as a rational function in the rates and in D , G . All coefficients of N and of the denominator must be nonnegative. The bound then holds for all positive rates and fuel potentials, independent of the rate range.

Trusted base. The verifier code (exact rational and symbolic arithmetic) is trusted. It passes a self-test with 18 of 18 cases, comprising true, false, boundary and rule-violation cases. Interval arithmetic is not part of the certificates reported here. Independent numerical re-searches by the verifier serve only as counter-checks.

Numerical candidates. Optimizer output, Pareto-front shapes and any value without a certificate (a) or (b) count only as numerical candidates and carry no result status.

The results were produced by an automated, verifier-gated laboratory in which every statement had to pass the code verifier before being kept.

4 Results

4.1 Main results

Theorem 1 (Hopfield chain, $n = 2$). *For `hopfield_n2`, $\eta \geq e^{-3\Delta}$ holds for all positive rates and all fuel potentials $\mu, \mu_P \geq 0$.*

Certificate. Certificate (b): $\eta - e^{-3\Delta} = N/D$ with 1374 and 2048 terms, all coefficients nonnegative (check time 6.8 s). Novelty status: not found in a targeted search of 40 abstracts on 2026-10-04. The novelty check classifies this as a reproduction of the Hopfield bound, proved here rigorously.

Theorem 2 (Proven bound for 50 topologies). *For 50 topologies of the verifier-generated family with at most two bound states (one unbound state, edge catalogue of the model), $\eta \geq 1/D^2$ holds for all positive rates and fuel potentials.*

Certificate. Certificate (b), symbolic: the bound is proved for 50 of 50 members. The complete count is consistent with the later classification. Novelty status: not found in a targeted search of 48 abstracts on 2026-10-04.

Theorem 3 (Exact violation in fam2₆₆). *For fam2_66 there exist rational rates in $[e^{-10}, e^{10}]$ satisfying local detailed balance with $\eta \leq 0.0001$ and $v \geq 0.0001$.*

Certificate. Certificate (a), exact: $\eta = 5.082802\text{e-}07$, $\sigma = 8.341626$ kT per product, $v = 1.4746\text{e-}03$. The edges of fam2_66 are binding at C0, fuel-driven conversion C0→C1, fuel-driven discard from C1, and product formation from C1. Novelty status: not found in a targeted search of 48 abstracts on 2026-10-04.

Theorem 4 (Violations in fam2₁₁ and fam2₉). *For each of the two topologies fam2_11 and fam2_9 there exist rational rates in $[e^{-10}, e^{10}]$ satisfying local detailed balance with $\eta \leq 0.0001$.*

Certificate. Certificate (a), exact for each case; 2 of 2 cases passed. Novelty status: not found in a targeted search of 42 abstracts on 2026-10-04.

Theorem 5 (Violations in ten topologies). *For each of ten topologies (fam2_11, fam2_35, fam2_39, fam2_41, fam2_43, fam2_45, fam2_58, fam2_67, fam2_75, fam2_9) there exist rational rates in $[e^{-10}, e^{10}]$ satisfying local detailed balance with $\eta \leq 0.0001$.*

Certificate. Certificate (a), exact for each case; 10 of 10 cases passed. Novelty status: not checked.

Theorem 6 (Consistency of the proven count). *The proven-bound count of 50 topologies with $\eta \geq 1/D^2$ is reproduced by the symbolic certificate (b) for 50 of 50 members.*

Certificate. Certificate (b) as in the theorem on 50 topologies. Novelty status: not checked.

Theorem 7 (Classification of the 88 topologies). *Among the 88 topologies with two bound states, the following holds:*

- 50 have the proven bound $\eta \geq e^{-2\Delta}$ (certificate (b));
- 11 have an exactly certified violation $\eta < e^{-2\Delta}$: fam2_9, fam2_11, fam2_35, fam2_39, fam2_41, fam2_43, fam2_45, fam2_58, fam2_66, fam2_67, fam2_75;
- 27 are open.

Certificate. Recomputed from the confirmed claims by the classification script `asd/klaskifikation.py`. Novelty status: the classification of the two-bound-state family and its counterexamples were not found in the literature search; editing mechanisms are qualitatively known.

Theorem 8 (Exact error rates of the violations). *All eleven violations are certified with rational rates in exact arithmetic, and all error rates are strictly below $e^{-2\Delta} = 1e-4$:*

Topology	η (exact)	Evidence
fam2__9	2.000856e-05	
fam2__11	4.417288e-05	
fam2__35	1.074798e-06	
fam2__39	1.010274e-06	
fam2__41	1.287193e-06	
fam2__43	1.386398e-12	
fam2__45	1.591024e-06	
fam2__58	7.272608e-08	
fam2__66	5.082802e-07	
fam2__67	4.098578e-07	
fam2__75	1.142234e-06	

Certificate. Certificate (a) for each case. Novelty status: as for the classification.

4.2 Supporting statements

Proposition 9 (Enumeration). *The generator yields 92 rule-compliant two-bound-state topologies, of which 4 are degenerate; 88 remain.*

Proposition 10 (Early symbolic subfamily). *For 3 topologies of the verifier-generated family with at most two bound states, $\eta \geq 1/D^2$ holds for all positive rates and fuel potentials (certificate (b), symbolic; 3 of 3 members). The same certificate was obtained for a search on whether sub- $e^{-2\Delta}$ error requires a fuel-driven discard exit. Novelty status: not found in targeted searches of 41 and 35 abstracts on 2026-10-04, respectively.*

Remark. This 3-member list is superseded by the complete list of the theorem on 50 topologies.

Proposition 11 (Verifier self-test). *The domain verifier passes its self-test in 18 of 18 cases.*

Numerical observation 12 (Novelty status). Against 158 checked findings, the chain bound is a reproduction, proved here rigorously; the reachable chain points are qualitatively known and exact here; the classification of the two-bound-state family and the counterexamples were not found in the search.

4.3 Examples (exactly certified points for the linear chain)

Example 13 ('hopfield_n1', *speedfloor*). There exist rational rates in $[e^{-10}, e^{10}]$ with local detailed balance such that simultaneously $\eta \leq 0.0004$, $\sigma \leq 6.0$ kT per product and $v \geq 0.001$. Certificate (a): $\eta = 3.650169e-04$, $\sigma \in [6.000000, 6.000000]$, $v = 1.0417e-01$. Novelty status: not found in a targeted search of 62 abstracts on 2026-10-04.

Example 14 ('hopfield_n1', *moderateerror*). There exist rational rates such that simultaneously $\eta \leq 0.01$ and $v \geq 0.001$. Certificate (a): $\eta = 5.909757e-03$, $\sigma = 3.998555$ kT , $v = 7.1214e-03$. Novelty status: not found in a targeted search of 45 abstracts on 2026-10-04.

Example 15 ('hopfield_n1', *lowdissipation*). There exist rational rates such that simultaneously $\eta \leq 0.00012$ and $\sigma \leq 0.2$ kT per product. Certificate (a): $\eta = 1.095823e-04$, $\sigma = 0.006292$ kT , $v = 9.1487e-09$. Novelty status: not found in a targeted search of 49 abstracts on 2026-10-04.

Example 16 (*‘hopfield_n1’, speedfloor, widedissipation*). There exist rational rates such that simultaneously $\eta \leq 0.0002$, $\sigma \leq 10 \text{ kT}$ per product and $v \geq 0.001$. Certificate (a): $\eta = 1.189200\text{e-}04$, $\sigma = 4.271813 \text{ kT}$, $v = 1.0000\text{e-}03$. Novelty status: not found in a targeted search of 37 abstracts on 2026-10-04.

Remark 17 (Status of computer-assisted parts). The proven bounds (the chain bound and the bound for 50 topologies) rest on symbolic positivity (certificate (b)) and are independent of the rate range. The violation statements rest on exact rational evaluation (certificate (a)) within the rate range of. A third party re-runs all certificates with the command given in the Declarations. Pareto-front shapes are not claimed here.

5 Negative results

- For 27 topologies, neither a proof of $\eta \geq e^{-2\Delta}$ nor an exact counterexample was obtained. These remain open.
- Failed verifier-gated attempts, with their reasons, are recorded in Appendix A.

6 Discussion, limitations and open questions

The linear chain with $n = 2$ obeys $\eta \geq e^{-3\Delta}$ for all rates. With two bound states, $e^{-2\Delta}$ is violated exactly by 11 of 88 topologies; the smallest certified error rate is $1.386398\text{e-}12$, for fam2_43. The violation in fam2_66 uses a fuel-driven discard edge from the product-forming state, with $\sigma = 8.341626 \text{ kT}$ per product and $v = 1.4746\text{e-}03$. All violation certificates use the rate range $[e^{-10}, e^{10}]$, whereas the proofs do not depend on it.

Open questions:

- What is the status of the 27 open topologies?
- Do all 11 violating topologies contain a fuel-driven discard edge? This is established here only for fam2_66.
- What are the minimal σ and the maximal v at which each violation occurs? Only η is tabulated for the topologies other than fam2_66.
- Do the chain bounds for the other chain lengths have symbolic certificates? Only $n = 2$ is certified here.

7 Declarations

Affiliation. Not specified.

AI usage. The results were produced and checked by an automated laboratory; every statement was verified by code.

Code and data availability. Repository <https://github.com/alizema700/Daddys-Project>, branch `claude/omnigent-lab`, directory `projects/omni_proofreading`. Reproduce all certificates with `python -m asd.recheck proofreading` and rebuild the paper with `python -m asd.paper --domain proofreading`.

Competing interests. None.

A Agentic laboratory

The laboratory ran 16 rounds with 14 verified statements and 2 negative results. Every round was preregistered before the experiment. Preregistration H6 (commit 6184672) preceded the first

run, and later additions are dated. The L2c success criterion (at least 5 counterexamples in one round) was missed: 2 in round 12, 3 in total. The prior literature search retrieved 2752 sources (arXiv, Europe PMC, Crossref, citation chain). Of these, 2502 were screened, and 158 findings had a word quote confirmed in the abstract by code. All 11 classic searches had hits. Round 12 verifier output: certificate (a) passed for 2 of 2 cases.

Agents worked in roles (sparing, numerics, skeptic, theorist, researcher). Their claims passed through the code verifier, and adversarial counter-checks were run against them. In total 29 counter-checks were made: 5 passed, 12 did not pass, and 12 were not executable.

Negative results.

- Question: does $\text{sub-}e^{-2\Delta}$ error in the two-bound-state family require a fuel-driven discard exit at the product state? No claim passed the verifier. All four `schränke_familie` attempts failed because of a verifier error that was fixed later.
- Question: further violations among the open topologies. The attempt with 11 cases passed certificate (a) for 10 of 11. `fam2_13` failed because a rate lay outside $[e^{-10}, e^{10}]$ ($|\log k| = 10.06$; derived reverse rates also count).

Pareto-front check. The minimal error rate is non-increasing in $\sigma_{\max}$, because the feasible set grows. A reported jump for $\sigma \geq 8$ in `hopfield_n1` is therefore an optimizer artefact.

Example 18 (`hopfield_n1`, *dissipationbudget*). There exist rational rates such that simultaneously $\eta \leq 0.0004$, $\sigma \leq 8 kT$ per product and $v \geq 0.001$. Certificates (a): $\eta = 3.156120\text{e-}04$, $\sigma = 6.000000 kT$, $v = 2.4255\text{e-}02$; and $\eta = 2.043712\text{e-}04$, $\sigma = 3.079633 kT$, $v = 2.2592\text{e-}03$. Novelty status: not found in a targeted search of 57 abstracts on 2026-10-04.

B Provenance

The provenance table mapping every statement to its evidence follows; it is generated automatically.

Statement	Evidence (claim ids)	Level
Section: Introduction	C-fakt-klassifikation, C-fakt-verletzungen-eta, C-proofreading-O14, C-proofreading-R11, C-proofreading-R12, C-proofreading-R4, C-proofreading-R9	computed_rigorous
Section: Model and assumptions	C-fakt-familie, C-modell	computed_rigorous
Section: Method	C-fakt-selbsttest, C-modell, C-proofreading-R4, C-proofreading-R9	computed_rigorous
Theorem 1 (Hopfield chain, $n = 2$)	C-proofreading-R9	computed_rigorous

Statement	Evidence (claim ids)	Level
Section: Main results	C-fakt-fam66, C-fakt-klassifikation, C-fakt-neuheit, C-fakt-verletzungen-eta, C-proofreading-O14, C-proofreading-O15, C-proofreading-R11, C-proofreading-R12, C-proofreading-R4, C-proofreading-R9	computed_rigorous, observed
Theorem 2 (Proven bound for 50 topologies)	C-proofreading-R11	computed_rigorous
Theorem 3 (Exact violation in fam2_66)	C-proofreading-R4	computed_rigorous
Theorem 4 (Violations in fam2_11 and fam2_9)	C-proofreading-R12	computed_rigorous
Theorem 5 (Violations in ten topologies)	C-proofreading-O14	computed_rigorous
Theorem 6 (Consistency of the proven count)	C-proofreading-O15	computed_rigorous
Theorem 7 (Classification of the 88 topologies)	C-fakt-klassifikation	computed_rigorous
Theorem 8 (Exact error rates of the violations)	C-fakt-verletzungen-eta	computed_rigorous
Proposition 1 (Enumeration)	C-fakt-familie	computed_rigorous
Proposition 2 (Early symbolic subfamily)	C-proofreading-R6, C-proofreading-R7	computed_rigorous
Section: Supporting statements	C-proofreading-R11	computed_rigorous
Proposition 3 (Verifier self-test)	C-fakt-selbsttest	computed_rigorous
Observation 1 (Novelty status)	C-fakt-neuheit	observed
Example 1 (hopfield_n1, speed floor)	C-proofreading-R1	computed_rigorous
Example 2 (hopfield_n1, moderate error)	C-proofreading-R3	computed_rigorous
Example 3 (hopfield_n1, low dissipation)	C-proofreading-R8	computed_rigorous
Example 4 (hopfield_n1, speed floor, wide dissipation)	C-proofreading-R10	computed_rigorous
Remark 1 (Status of computer-assisted parts)	C-modell, C-proofreading-R11, C-proofreading-R9, C-verfuegbarkeit	computed_rigorous, observed
Section: Negative results	C-fakt-klassifikation	computed_rigorous
Section: Discussion, limitations and open questions	C-fakt-fam66, C-fakt-klassifikation, C-fakt-verletzungen-eta, C-modell, C-proofreading-R9	computed_rigorous
Section: Declarations	C-verfuegbarkeit	observed

Statement	Evidence (claim ids)	Level
Section: Appendix A: Agentic laboratory	C-fakt-praereg, C-fakt-r12, C-fakt-recherche, C-methode, C-neg1, C-neg2, C-proofreading-R1-RT1, C-proofreading-R1-RT2, C-redteam	computed_rigorous, observed
Example 5 (hopfield_n1, dissipation budget)	C-proofreading-R2	computed_rigorous
Section: Appendix C: Verifier self-test	C-fakt-selbsttest, C-proofreading-O14, C-proofreading-O15, C-proofreading-R1, C-proofreading-R10, C-proofreading-R11, C-proofreading-R12, C-proofreading-R2, C-proofreading-R4, C-proofreading-R6, C-proofreading-R7, C-proofreading-R8, C-proofreading-R9	computed_rigorous

The complete automatic checking protocol of the writing process is stored as `pruefprotokoll.json` in the project directory of the repository.

C Verifier self-test

The self-test of the domain verifier covers true statements, false statements, boundary cases and rule violations. The verifier passes 18 of 18 cases.

References

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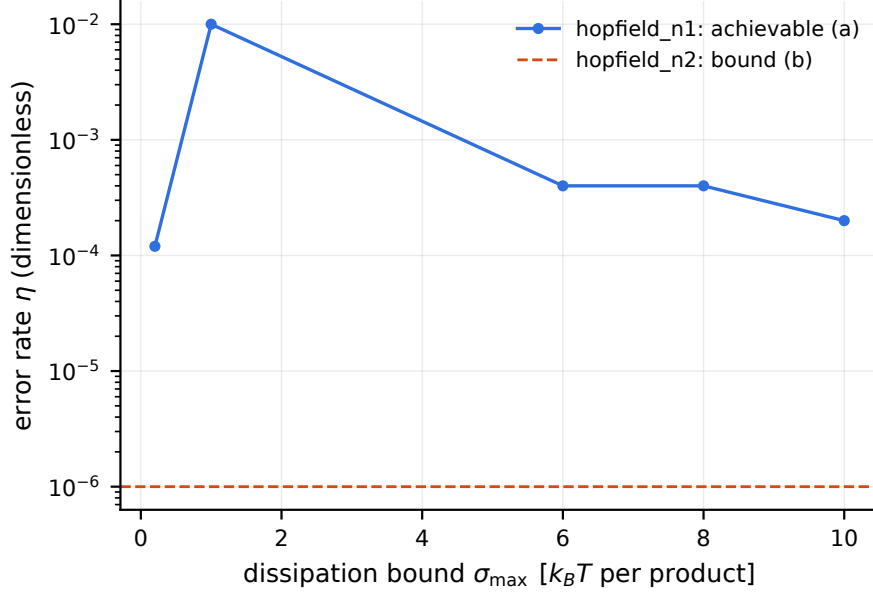


Figure 1: Certified achievable error thresholds (dots; each dot is the threshold of one exact certificate, not an optimum) and proven lower bounds (dashed lines).

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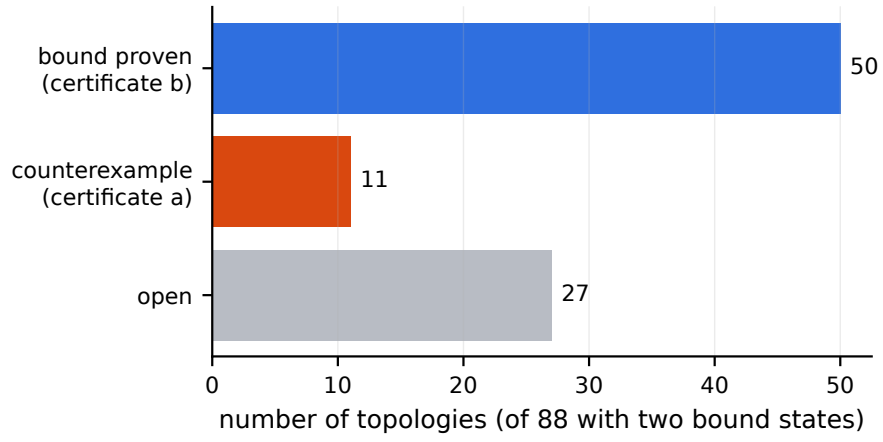


Figure 2: Classification of the 88 non-degenerate networks with two bound states with respect to the bound $\eta \geq e^{-2\Delta}$.